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A lateral transshipment model for perishable inventory management

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Since inventory costs account for half of logistics costs, optimal inventory management to minimise total inventory costs remains a sustainable competitive advantage. Lateral transshipment (LT) is evidently a proven strategy to minimise total inventory costs. The additional LT costs are more than compensated by lowering the stock-out costs. Previous LT models have not been applied to perishable products. Our proposed LT model embodies spoilage costs in the total inventory costs function with the other cost components (purchase from a regular supplier, LT, backordering and holding), and optimises the trade-off among these five key cost components. Numerical examples from a supermarket chain case study demonstrate that, as compared against the no or lower spoilage costs scenarios, lower LT costs are required to trigger the decision point for implementing LT in the higher spoilage costs scenario. However, common to both the with and without spoilage costs scenarios, LT is still the preferred strategy to minimise total inventory costs, given the decision rules are satisfied.

Keywords: lateral transshipment; perishable inventory management; spoilage costs; heuristic decision rule

1. Introduction

In response to changes in consumer preference for fresh food, supermarket chains adjust their retailing formats, highlighting the prominence of the vegetable department as a centrepiece of their operations, for example, as espoused by the Australian supermarket chain Woolworth's 'Fresh Food People' as an operations principle. Daily supply of fresh food turns into a major strategy to attract customers. Fresh foods are increasingly becoming a primary decision variable for consumer deciding to shop at one supermarket instead of another. However, management of perishable products remains a real challenge for supermarket chains and grocery retailers due to the high level of spoilage and damage, which can result in financial loss either through a substantial marked down near expiry of the shelf life or mere disposal. The inventory control problems associated with perishable or short life cycle product are commonly found in many other industries such as blood, fresh and packaged foods, flowers, designer fashion clothing, airline and hotel, entertainment, publishing, electronics and pharmaceutical industries as welll (Trappey and Wu 2008).

As market competition intensifies, manufacturers/wholesalers are increasingly focusing on cost efficiency and market responsive strategies. Since inventory costs account for half of logistics costs (Establish 2013), optimal inventory management to minimise total inventory costs remains as an effective strategy for a sustainable competitive advantage. Conserving a low inventory level just enough for instantaneous availability for use or sales purpose has thus been adopted. However, the accompanying higher risk of stock outages is breaching the minimal customer service standard and effect the consequential risk of sales losses and customer dissatisfaction. Needham and Evers (1998) use 50% of the item cost to represent their initially estimated stock-out cost, but comment that the stock-out cost penalty would exceed the item cost due to the potential loss of future sales and after sales service support. To manage these adversities, manufacturers/wholesalers usually increase the flexibility of inventory systems by adopting a mix of fast response lateral transshipment (LT) from other warehouses and simultaneously backordering from their usual suppliers to match the stochastic demand. Needham and Evers (1998) prefer to use the range of 50–800% of the item cost in their stock-out cost estimation, which is higher than the transshipment costs. Decision rules on LT that support this decision-making process have practical value for inventory management practitioners. An earlier research study (Lau and Nakandala 2012) has formulated decision rules for determining whether LT is preferred to effect minimisation of the total inventory costs, however, only applicable to non-perishable products.

Previous research investigating the use of LT in the fresh food supply chains has been very limited. Perishability of fresh food products affect product quality, price and waste based on the rate of degradation of quality. If the LT

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decisions in the logistics practices do not make allowances for the downgrading of quality of fresh food products to reflect the unique features and requirements of the fresh food supply chain, then the total inventory costs for these perishable products would turn out to be higher than the optimised level. Our case study is based on a supermarket chain that we provided consulting service in relation to the practice of LT logistics. Using the previous LT model (Lau and Nakandala 2012) to make LT-related decisions did not generate the expected benefits. Upon investigation of the products and operations of this supermarket chain, our research identified the importance of considering the perishability of products as an influencing variable in the total inventory cost function. This study aims to develop a decision method to assist logistics practitioners in the fresh food supply chains to know if adopting LT in inventory management is cost effective.

This study is novel as, to the best of the authors' knowledge, there have not been any prior studies investigating LT in the fresh food industry. The authors' previous work on LT for non-perishable goods was enhanced significantly to consider the effects of food spoilage cost on the development of total inventory costs model and decision rules to assist cost-optimised decisions for LT. However, this new approach does not undermine previous scholarly work, but build upon previous constructs to develop a more pertinent decision model to achieve more cost-effective inventory management for the fresh food SCM environment. The proposed LT approach for perishable goods is validated in this large supermarket chain case study company.

Section 2 reviews the relevant literature, the decision rules derived from the proposed mathematical model with application to the transshipment decision-making process through a case study have practical values and implications for management. Section 5 concludes with a discussion of the effectiveness and limitations of the decision model, with suggestions for further research.

2. Literature review

Previous research studies on managing perishable inventories identify the impacts of demand uncertainty, lead times, seasonality of the product and the requirements for innovative inventory management policies (Nahmias 2011; Zhou, Leung, and Pierskalla 2011). For example, an age-based replenishment policy was proposed by Duan and Liao (2013) by considering the age distribution of the items in replenishments. The EWA policy developed by Broekmeulen and van Donselaar (2009) takes an approach to correct the inventory position considering the estimated amount of outdating of items. A mixed mode of replenishment was proposed by Zhou, Leung, and Pierskalla (2011) in a study on cost-optimised inventory policies for platelets and blood in a hospital context; they developed a methodology to determine whether the hospital should order daily or adopt a combined strategy of ordering every other day with optional expedited orders is cost effective. They identified the complexity associated with optimising the combination of regular and expedited orders to minimise the total inventory costs. Expedited orders to manage the emergencies or as a supplement to regular orders from regular suppliers have been the dominant approach in previous studies (Fontaine et al. 2009; Zhou, Leung, and Pierskalla 2011) and LT as an alternative has received a limited attention.

Analytical method has traditionally been adopted to assess and quantify system behaviours associated with LT models. The analytical LT models have progressively been advanced in the literature (Archibald 2007; Wong, Cattrysse, and Oudheusden 2005; Wong et al. 2006). However, the added structural complexities in the LT research become too cumbersome for developing the optimal solutions, especially when the demand and supply functions are modelled as stochastic processes. To compensate for the mathematically intractable exact analysis, search procedures are implemented to discover the optimal solutionse. Simulation is one proven approach to generate numerical systems representation of the complex LT models for assessing interactions of system components and their causal relationships, as well as evaluating the optimal performance of various competing models. However, publication of simulation studies in the LT literature is limited, though logistics visionaries have expressed a consensual overwhelming call to promote simulation to supplement the traditional analytical method (Datta and Christopher 2011; Davis-Sramek and Fugate 2007; Hayya et al. 2006). As a response, results of the analytical models of our study are validated by Monte Carlo simulation.

In the LT literature, two major types of transshipments are identified on the basis of timing of the transshipments (Paterson et al. 2011; Seidscher and Minner 2013; Wong et al. 2006), namely the proactive and reactive transshipments. Proactive LT is a preventive approach. Before an inventory shortage occurs, the LT decision is implemented at a predetermined point of time to avoid future stock-out. As a result, inventory levels of different locations at the same echelon are rebalanced to reduce the risk of future stock-out (Bertrand and Bookbinder 1998; Jönsson and Silver 1987; Tagaras and Vlachos 2002). The alternative approach is reactive LT (Chiu and Huang 2003), which responds to an occurrence of inventory shortage as demand has been realised. As a result, location with surplus stock on hand transfers inventory to another location with stock deficiency (Krishnan and Rao 1965; Robinson 1990). This study concentrates on developing a discrete-time reactive LT model to optimise total inventory costs.

In recent years, researchers start to examine LT in a virtual e-Commerce environment. Customers can benefit from the advantage of the online platform, e.g. Amazon.com, to gain instant direct access to a wide range of product choice and information, price comparison, minimal stock-out, however, at the expense of longer delivery lead time. As a strategic response, large 'bricks and mortar' retail chains and companies today also expand into their online sales channels, and therefore also face similar challenges in designing their distribution structure to accommodate both direct from own warehouses and suppliers. Apart from providing customers with the normal benefits through the retail channel, e.g. touch and feel experience of the products and services, the dual channel also reduces the chance of sales return in failed product and poor delivery quality (Chen, Kaya, and Ozer 2008; Huang, Yang, and Zhang 2012). Furthermore, the order fulfilment in the dual-channel SCM is usually supported by drop shipping with the advantages of risk pooling to reduce the costs of inventory, transportation and stock-out.

He, Zhang, and Yao (2014) apply unidirectional transshipment under the two-echelon dual-channel model and find that the transshipment price mechanism always coordinates the supply chain. Zhao et al. (2015) design a new online-to-offline (OTO) business model to enhance the traditional dual-channel supply chain and drop-shipping model to take advantage of LT to fulfil customer order from inventories stored at the nearest retailer's warehouse. This LT approach can reduce the stock-out risk, improve customer service level with higher fill rate, minimal transshipment costs and lower aggregate inventory costs (Belgasmi, Ben Saïd, and Ghédira 2008, 2013). Using a large survey sample of 71,000 customers, Gruen and Corsten (2004) discovers that consumers have little patience for stock-outs, with 85% choose substitute or walk out to other stores when cannot find precise products on shelf. Under this circumstance, reactive LT fails to accommodate this type of consumer behaviours. Therefore, leading fashion chains like Zara and H&M adopt preventive LT to minimise stock-outs (Caro et al. 2010). As such, our study focuses on implementing transshipment in the dual-channel and OTO business model of the case study supermarket chain company to achieve the desired outcomes of minimising total inventory costs when customers have little patience for stock-outs.

Through literature review, a research gap of lacking LT implementation in the perishable inventory management environment is identified. Perishables differ from non-perishables in terms of shelf life, which is measured from the day the product is produced until its expiry date, when it becomes obsolete or unsuitable for consumption. Perishable products, e.g. blood, fresh and packaged foods, fashion, electronics, drugs, film, plants maintain a stable value until the end of shelf life or expiration date, either deterministic or stochastic. In the perishable inventory management literature, a key research issue is to find the optimal balance between product availability and expiry of shelf life (van Donselaar and Broekmeulen 2012; Kim, Wu, and Huang 2015; Trappey and Wu 2008). Though LT is now an entrenched research field (Paterson et al. 2011; Seidscher and Minner 2013; Wong et al. 2006), and previous studies have demonstrated the benefits of LT to improve operational efficiencies through inventory reallocation, nevertheless, investigations of LT on perishable products with limited shelf life are surprisingly sporadic.

In our study, a major supermarket chain owns and centrally manages a large number of physical and virtual retail stores, served by five distribution centres with LT to supplement order from suppliers for replenishment of perishable and non-perishable products. As emphasised by the senior management, an LT decision support model to optimise total inventory costs has high potential value in managing the perishable inventory systems. Our study aims to develop a discrete-time LT model to investigate the possibility of using LT to optimise the total costs of perishable inventory.

3. Model development

Previous research investigating the use of LT in the fresh food supply chains has been very limited. Perishability of fresh food products affect product quality, price and waste based on the rate of degradation of quality. This study is a novel attempt to develop a decision method to assist logistics practitioners in the fresh food supply chains to know if adopting LT in inventory management is cost effective. The development of the model was informed by our previous work in this area (Lau and Nakandala 2012), but there has been substantial work undertaken to adapt the model to the fresh food supply chains and introduce the spoilage cost as a cost element in our model.

Our case study is based on a firm that we provided consulting service in relation to the practice of LT. Using our original LT model to make LT-related decisions did not generate the expected benefits. We investigated the products and operations of this firm and identified the importance of considering the perishability of products as an influencing variable in the total inventory cost model. In this paper, the previous LT model has been extensively revised to reflect the unique features and requirements of the fresh food supply chain, taking into account the downgrading of quality of fresh food products. After modifying our LT model to introduce the spoilage costs, the results demonstrated that perishable products (unlike normal products) with spoilage costs require a lower LT cost to make the LT decision profitable to fulfil urgent outstanding orders. If this principle is neglected, then the total inventory costs for these perishable products would turn out to be higher than the optimised level.

A supermarket chain maintains a limited inventory in all its retail stores to stay cost competitive. To fulfil retailer demand and avoid inventory shortage require constantly sourcing from external suppliers. There are a finite number of distribution centres operating autonomously in a local region. Following Axsäter (2006) with the assumption that the retailer demand from these distribution centres follows a compound Poisson distribution. The distribution centres usually use their suppliers to replenish their stocks, within a promised delivery lead time. They may also opt to source laterally from other distribution centres. For model simplicity, we assume instant transshipment of goods due to their close proximity. Such transactions among distribution centres are called LT, and they are usually more expensive than the usual purchase price charged by suppliers (Axsäter 2006).

All the distribution centres apply a periodic review policy, as described in Rosenshine and Obee (1976), to regular replenishments from suppliers. The central distribution unit reviews the unsatisfied demands in the previous period, the inventory position and the expected demand in the current period to plan for the orders to be placed at the beginning of this period. Unsatisfied demands or surplus orders in this period will be counted as initial demands or added to the inventory position of the next period. If the distribution centre does not employ a (R, Q) review policy, its inventory position could drop below zero, which potentially increases the backordering costs that outweigh reduction in the holding costs of the distribution centre. One possible solution is to make transshipment from other distribution centres so as to restore the inventory position of the distribution centre to a non-negative level as soon as possible.

The input parameters for computing the total inventory costs consist of:

(1) Purchasing costs include all the costs associated with the purchasing process that include planning order, requisition, monitoring and controlling the order progress, transportation and shipping, receiving, inspection, handling and storage, accounting and auditing costs. Costs involve labour, equipment and related resources to perform purchasing process.
(2) Backordering is opted when stock on hand is not adequate to meet customer demand. Hence, backordering costs include the cost of backordering from suppliers, lost sales due to stock unavailability and future costs such as loss of future sales and customer loyalty due to customer dissatisfaction. Late delivery payments and penalties according contractual obligations are also included in backordering costs. However, it is largely resorted to judgement and thus generally ignored in inventory costing due to its estimation uncertainty.
(3) Holding costs consider the costs of holding items in stock and include interest on loans to finance inventory or opportunity costs of the investment on inventory. Costs associated with holding inventory include rent, provision of facilities, heating, cooling, lighting, security, refrigeration, administrative, handling and storage, transportation, insurance and taxes. It amounts to approximately 15 to 30% of the total inventory costs (Christopher 2016), but is difficult to calculate with a high degree of accuracy, and is often underestimated.
(4) Spoilage costs include loss in product value over time until worthless. When perishable goods such as blood, fresh and packaged foods, flowers, designer fashion clothing, airline seat or hotel room, tickets to football game or concert, magazines, electronics, drugs, film, plants are unsold when expire with short shelf life. For grocery foods, the shelf life can further be shortened and product quality level can deteriorate by inadequate temperature control, storage and handling as perishable products traverse the supply chain, resulting in possible product shrinkage, damage or spoilage. Additionally, food waste disposal can be a costly environmental burden.

3.1 *Proposed LT model with perishable products*

The following notations and assumptions form the basis for building the mathematical model of the LT problem with perishable products. We consider that the ith wholesaler, W_i has the option of ordering from the jth regular supplier, S_{ij} and laterally transship from another kth wholesaler, W_k.

W_i = the ith distribution centre,
N_i = the total number of suppliers to the distribution centre W_i,
S_{ij} = the jth supplier of the distribution centre W_i,
p_{ij} = the unit selling price by S_{ij} to W_i,
q_{ik} = the unit cost of intra-shipment for W_i to intraship from W_k,
b_i = unit back-order cost at W_i per unit time,
h_i = unit holding cost for W_i per unit time,
t_0 = start of the scheduling period, $t = 0$,
$g_{ij}(t)$ = delivery lead time probability mass function of S_{ij},

L_{ij} = lead time of S_{ij} with duration equal to L_{ij} times unit time interval,
$L_{ij}^{\max}$ = the maximal lead time of S_{ij},
$d_i(0)$ = the initial retailer demand at t = 0 appearing at W_i,
$\lambda_i(t)$ = the retailer arrival intensity during the tth time interval at W_i,
$f_{\lambda,m}^{n}$ = the probability of n retailers arriving at W_i with a total demand of m,
$\hat{d}_i(t)$ = the expected retailer demand at distribution centre W_i in the tth time interval,
$\hat{D}_{ij}$ = the expected retailer demand at distribution centre W_i over $L_{ij}^{\max}$

To determine the optimal quantity and timing of LT, the four key cost components, purchasing cost, backordering cost, holding cost and spoilage cost of the total inventory costs should be minimised.

$$C_{ijk}(x) = P_{ijk}(x) + B_{ijk}(x) + H_{ijk}(x) + S_{ijk}(x). \tag{1}$$

There appear penalties for ordering too much, e.g. outdating, marginal ordering cost, holding cost and opportunity cost for carrying stock on hand until end of the period. On the other hand, there are also penalties for ordering too little or excess demand, e.g. backordering costs or lost sales (Nahmias 2011). Our study develops a model that optimises the total inventory costs, given the interactions of these four key cost components, within the context of grocery store environment where LT can be utilised to expressively reduce the total inventory costs. The mathematical models for these four key components will be formulated in the following sub-sections.

Considering that the supplier, S_{ij} of the wholesaler, W_i takes L_{ij} lead time to deliver the order, the expected lead time of $E(L_{ij})$ s given by $E(L_{ij}) =_0^{L_{ij}} tg_{ij}(t)\mathrm{d}t$ and the expected demand during the maximum lead time of $L_{ij}^{\max}$ is given by $\hat{D}_{ij} = \int_{k=1}^{k=L_{ij}^{\max}} \hat{d}(k)$. Based on Lau and Nakandala (2012), the demand is then determined by.

$$\hat{D}_{ij} = \sum_{k=1}^{L_{ij}^{\max}} e^{-\lambda k} \sum_{m=1}^{+\infty} \sum_{n=1}^{m} \frac{\{\lambda_i k\}^n}{n!} m f_{1,m}^{n}(n)$$

3.1.1 *The purchasing, backordering and holding cost components*

Purchasing costs consist of two components, depending on the source of supply, i.e. either from a regular supplier or another distribution centre via LT. The purchasing cost is the total of the difference between the outstanding demand and the inventory level at the beginning of the scheduling period ($t = 0$), the laterally transshipped quantity and the expected demand over $L_{ij}^{\max}$, i.e.

$$P_{ijk}(x) = p_{ij}(d_i(0) - l_i(0) + \hat{D}_{ij} - x) + q_{ik}(x) \tag{2}$$

For the backordering costs, we assume that any unfulfilled retailer demands would be backordered as in Axsäter (2003). When the retailer orders cannot be fulfilled immediately, backordering costs would be incurred until the next shipment arrives (Nahmias 2009). The initial back-ordered quantity is $(d_i(0) - l_i(0) - x)$, with the back-ordering time equal to the expected order arrival time. Assume that transshipping cost is significantly higher than the purchasing cost, under normal logistics operations, orders are sourced from suppliers. Only the urgent orders would be transshipped. We also assume that transshipped orders have zero lead time. At the start of the scheduling period, any outstanding order would generate backordering costs. If retailer demands would not be fulfilled during the time range $0 \le t \le \{E(L)_{ij} - 1\}$, the backordering costs aggravate.

Based on the expected retailer demand, the following Table 1 defines the back-order time for the unfulfilled demand.

Table 1. Unfulfilled demand for the backorder time periods.

Time period of demand	Demand	Backorder period
at t_0	$(d_i(0) - l_i(0) - x$	$E(L_{ij})$
$[t_0 - t_1]$	$\hat{d}_i(1)$	$E(L_{ij}) - 1$
$[t_1 - t_2]$	$\hat{d}_i(2) - \hat{d}_i(1)$	$E(L_{ij}) - 2$
$[t_2 - t_3]$	$\hat{d}_i(3) - \hat{d}_i(2)$	$E(L_{ij}) - 3$
…	…	…
$\left[\left\{E(L_{ij}) - 2\right\} - \left\{E(L_{ij}) - 3\right\}\right]$	$\hat{d}_i(E(L_{ij} - 2) - \hat{d}_i(E(L_{ij} - 3)$	2
$\left[\left\{E(L_{ij}) - 1\right\} - \left\{E(L_{ij}) - 2\right\}\right]$	$\hat{d}_i(E(L_{ij} - 1) - \hat{d}_i(E(L_{ij} - 2)$	1

Sourced: Lau and Nakandala (2012).

As such, the backorder cost B_{ijk} can be formulated as follows:

$$B_{ijk}(x) = b_{ij}(d_i(0) - l_i(0) - x)E(L_{ij}) + b_{ij}\int_{t=1}^{E(L_{ij})-1}\{\hat{d}(t) - \hat{d}(t-1)\}\{E(L_{ij}) - t\}\mathrm{d}t \quad (3)$$

For retailer demands that arrive after the expected supplier lead time, no backordering costs would be triggered. However, the increase in stock will generate holding costs.

Holding costs are the carrying costs of keeping inventory, with cost objects including the physical storage space, taxes and insurance, breakage and other damages and opportunity costs of alternative investment (Nahmias 2009). Assume retailer demand as a compound Poisson process and λ the retailer's arrival intensity, as formulated by Axsäter (2003), Table 2 presents the holding period for the stock received from the supplier at $E(L_{ij})$.

The total holding costs H_{ijk} can be formulated as follows:

$$H_{ijk}(x) = \int_{t=E(L_{ij})+1}^{L_{ij}^{\max}}\left\{\hat{d}(t) - \hat{d}(t-1)\right\}\{t - E(L_{ij})\}h_{ij}\mathrm{d}t \quad (4)$$

3.1.2 *Spoilage costs*

Due to excessive inventories and/or inferior quality control, perishable products that are not sold to consumers before their expiry date or spoilage are either marked down before their shelf life or disposed of thereafter and the discarding costs are incurred (Wang and Li 2012; Yu and Nagurney 2013). The spoilage or wastage costs are estimated to cost billions of dollars financial loss for the European grocery sector each year (Kärkkäinen 2003).

Let S_{ijk} be the spoilage costs for the distribution centre W_i after the perishable products expire their fixed shelf life L_{ijk}^{msl}, where *msl* is the maximum shelf life, and $E(w_{ijk})$ be the expected unit spoilage cost. However, when comparing between the LT and the supplier order options, it is the $(q_{ik} - p_{ij})$ component that should be taken into account because this cost differential represents the transportation and administration costs associated with the LT, i.e.

$$S_{ijk}(x) = E(w_{ijk})(q_{ik} - p_{ij})\left[(l_i(0) + x - d_i(0)) - \int_{t=1}^{E(L_{ijk}^{msl})}\left\{\hat{d}(t) - \hat{d}(t-1)\right\}\mathrm{d}t\right] \quad (5)$$

In other words, the p_{ij} component is the opportunity cost of that company when deciding whether to transship or not. Since the expected spoilage cost $E(w_{ijk})$ is always non-negative, the gamma distribution is a more appropriate representation than the Gaussian distribution (Dolgui and Proth 2010). The mean $\theta = \alpha\beta$, where α and β are positive parameters. α and β can be estimated by the mean and standard deviation of the panel time series of the spoilage costs. Since the mean is $\alpha\beta$ and the variance is $\alpha\beta^2$, therefore β = variance/mean; and $\alpha = \text{mean}/\beta$. And the spoilage costs S_{ijk} can be simplified as follows:

$$S_{ijk}(x) = \theta\left[(p_{ij} - q_{ik})(d_i(0) - l_i(0) + \hat{D}_{ij}^{msl} - x)\right] \quad (6)$$

where $\hat{D}_{ij}^{msl} = \int_{t=1}^{E(L_{ijk}^{msl})}\{\hat{d}(t) - \hat{d}(t-1)\}$ is the estimated demand until the shelf life of the perishable product expires.

Table 2. Holding period of the stock received from the supplier.

Time period of demand	Demand	Holding period
$\{E(L_{ij})+1\} - E(L_{ij})$	$\hat{d}_i(E(L_{ij})+1) - \hat{d}_iE(L_{ij})$	1
$\{E(L_{ij})+2\} - \{E(L_{ij})+1\}$	$\hat{d}_i(E(L_{ij})+2) - \hat{d}_i(E(L_{ij})+1)$	2
$\{E(L_{ij})+3\} - \{E(L_{ij})+2\}$	$\hat{d}_i(E(L_{ij})+3) - \hat{d}_i(E(L_{ij})+2)$	3
…	…	…
$\left\{L_{ij}^{\max}-1\right\} - \left\{L_{ij}^{\max}-2\right\}$	$\hat{d}_i(L_{ij}^{\max}-1) - \hat{d}_i(L_{ij}^{\max}-2)$	$E(L_{ij}^{\max}-1) - E(L_{ij})$
$L_{ij}^{\max} - \left\{L_{ij}^{\max}-1\right\}$	$\hat{d}_i(L_{ij}^{\max}) - \hat{d}_i(L_{ij}^{\max}-1)$	$E(L_{ij}^{\max}) - E(L_{ij})$

Source: Lau and Nakandala (2012).

Though the parameter θ can be estimated via the Kaplan–Meier semi-parametric proportional hazards model and nonparametric estimator (Kaplan and Meier 1958), in practice, θ is usually measured from actual sales and purchase data or managerial judgement. The value of θ is estimated to range from 20 to 60% in agricultural fresh produce that are wasted in the supply chain (Widodo et al. 2006), and 15 to 20% due to damage and spoilage for perishable food loss at grocery retailers (Ferguson and Ketzenberg 2006; Shrady 1995). This proposed spoilage ratio can accommodate the three categories of perishable inventory models as observed by Bakker, Riezebos, and Teunter (2012) in the perishable inventory management literature. Furthermore, temperature is a critical control element in the three categories of perishable inventory models in determining the quality level of the perishable products, i.e. the value of θ. For example, Kouki et al. (2013) emphasise that the fixed lifetime category implicitly assumes that packaging of 'use by/consume by' label must be maintained under appropriate time and temperature conditions that reasonably as expected during transportation and storage. In the literature on perishable products with fixed shelf life, especially for fresh produce items, the triggering of spoilage varies significantly with different temperatures (Blackburn and Scudder 2009; Rong, Akkerman, and Grunow 2011). Although investments in technology may increase shelf life duration of perishable products, evaluation must be carefully conducted to justify such investments economically (Avinadav, Herbon, and Spiegel 2013). On the other hand, supermarket chains have, in practice, strict operational guidelines to specify a temperature range so as to maximise the shelf life of fresh produce items. Therefore, embodying temperature as an influential factor on the quality level of perishable products and the value of θ is not explored in our study.

3.2 *Decision rule for transshipment orders*

Substituting the functions of the purchasing, backordering, holding and spoilage costs (Equations (2)–(4), and (6)) into total inventory costs function (Equation (1)).

$$C_{ijk}(x) = \left[(1+\theta)p_{ij} - \theta q_{ik}\right](d_i(0) - l_i(0) - x) + p_{ij}\hat{D}_{ij} + \theta(p_{ij} - q_{ik})\hat{D}_{ij}^{msl} + q_{ik}(x) + b_{ij}(d_i(0) - l_i(0) - x)E(L_{ij}) \\ + b_{ij}\int_{t=1}^{E(L_{ij})-1}\{\hat{d}(t) - \hat{d}(t-1)\}\{E(L_{ij}) - t\}\mathrm{d}t + \int_{t=E(L_{ij})+1}^{L_{ij}^{\max}}\{\hat{d}(t) - \hat{d}(t-1)\}\{t - E(L_{ij})\}h_{ij}\mathrm{d}t \quad (7)$$

Alternatively,

$$C_{ijk}(x) = \left[-(1+\theta)p_{ij} - E(L_{ij})b_i + (1+\theta)q_{ik}\right]x + \left[(1+\theta)p_{ij} - \theta q_{ik} + E(L_{ij})b_i\right](d_i(0) - l_i(0)) + p_{ij}\hat{D}_{ij} + \theta(p_{ij} \\ - q_{ik})\hat{D}_{ij}^{msl} + b_{ij}\int_{t=1}^{E(L_{ij})-1}\{\hat{d}(t) - \hat{d}(t-1)\}\{E(L_{ij}) - t\}\mathrm{d}t + \int_{t=E(L_{ij})+1}^{L_{ij}^{\max}}\{\hat{d}(t) - \hat{d}(t-1)\}\{t - E(L_{ij})\}h_{ij}\mathrm{d}t \quad (8)$$

This total inventory costs function (Equation (8)) is linear with respect to the quantity of transshipment x. When the tangent of this linear function, i.e. $-(1+\theta)p_{ij} - E(L_{ij})b_i + (1+\theta)q_{ik}$, is negative, the total inventory costs decrease with increasing quantity transshipped.

Hence, the decision rule for transshipments is

$$-(1+\theta)p_{ij} - E(L_{ij})b_i + (1+\theta)q_{ik} < 0 \quad (9)$$

Alternatively, when the condition $q_{ik} < \left(p_{ij} + \frac{E(L_{ij})b_i}{(1+\theta)}\right)$ is satisfied, the higher the quantity transshipped, the lower the total inventory costs. In other words, the logistics decision to fulfil the demand by ordering from the supplier S_{ij} would be made if the above decision rule is not satisfied. In the scenario of no spoilage costs (Lau and Nakandala 2012), the decision rule is $q_{ik} < p_{ij} + (L_{ij})b_i$, i.e. the backordering costs $(L_{ij})b_i$ would not be reduced by a factor of $(1+\theta)$ that embodies the additional spoilage for perishable products. In comparison with the no spoilage costs scenario, the right-hand side of the decision rule for the spoilage costs scenario $q_{ik} < \left(p_{ij} + \frac{E(L_{ij})b_i}{(1+\theta)}\right)$ is relatively smaller since θ is always positive, and thus requiring the costs of LT q_{ik} at a lower level to trigger LT.

4. Case study

Our research approach utilises the LT model to assist managerial supply chain practitioners to improve their supply chain systems performance by minimising total inventory costs that embody perishable waste as one key cost component. The relevant input indicators have been identified when designing the roadmap for implementing LT (Lau, Nakandala, and Shum 2016). Data analysis reveals the baseline and the problems associated with the no LT and perishable costs scenario. Tapping into the literature, expert opinions and management experiences, various interventions and evaluation criteria have been investigated in a comparative study to determine why and what intervention is most suitable to improve inventory management. Relevant results from our study will be summarised to facilitate implementation of LT as part of the perishable inventory management systems. This practical solution offers managerial insights and can easily be generalised to other companies to achieve similar operational benefits.

This case study company is a major supermarket chain in Australia, with one national and seven regional distribution centres in each of the seven major cities in Australia to serve a diverse range of retail stores across wide geographies. Since most POS systems in its retail stores have real-time access to sales and inventory data, continuous review policy seems feasible and preferred. However, certain limitations are violating the other conditions for continuous review policy, making periodic review policy a necessity, including pre-determined schedule, fixed contracts confirmed with customers and shipping companies, simultaneous delivery of a variety of goods, batch update in ERP inventory databases and inventory decisions are made as per predefined cycles. Therefore, it is more appropriate for this case study company to adopt a periodic review policy.

The objective of our research is to measure current perishable inventory management performance and improve its performance by implementing the decision rules of LT so as to minimise total inventory costs. The key decision is the optimal division of inventory between the central warehouse and among retail stores. Higher customer service level can be achieved when more inventories are positioned at retail stores, but with the associated increase in transportation, holding and spoilage costs. Thus, an optimal balance is central to minimising the cost objective. The advantage of positioning more inventories in the national and regional distribution centres is risk pooling that reduces the total systems inventory costs. However, this configuration can cause shipment delay that may adversely impact on customer service level. Moreover, it is not an easy step to restore subsequent inventory imbalances across the regional distribution centres and retail stores if LT is not part of the normal replenishment process.

In apply these LT decision rules derived in Section 4.1, the parameter inputs of the retailer arrival intensity (λ), backordering costs (b_1), unit holding costs (h_1) and spoilage ratio are measured from historical retailer orders, sales, procurement and inventory accounting records of this supermarket chain. Taking one perishable product item for illustration, with $\lambda = 3$, $h_1 = 2$, $b_1 = 2$, $\theta = 0.1$. Since the retailer demand is essentially finite, the upper bound of the retailer demand over the scheduling period at the regional distribution centre $\hat{D}_{ij}^{msl} = 500$ units, and the retail demand over the maximum shelf life is $\hat{D}_{ij}^{msl} = 400$ units.

4.1 *Case – One supplier and one other regional distribution centre*

First, let's consider a situation where the regional distribution centre, W_1 has only one other regional distribution centre, W_2 and one supplier, S_1. Assume the maximum lead time of $L_{11}^{\max} = 10$, the unit purchasing cost from the S_1 is $p_{11} = 2.2$, and the inventory level of this regional distribution centre at $t = 0$ is $l_1(0) = 0$ and the initial demand at $t = 0$ is $d_1(0) = 6$. Using four different configurations of LT costs q and expected supplier lead time $E(L)$, the decision rule $(1+\theta)(q_{12} - p_{11}) < (E(L_{11})b_1$ is tested to determine whether LT should be implemented or not, as shown in Table 3.

For the costing scenarios of C_{111} and C_{133}, the backordering costs are higher than the spoilage adjusted LT price differential, thus the decision rule is satisfied. Therefore, LT is implemented, with quantity transshipped from the other regional distribution centre W_2 to minimise the total inventory costs.

Tables 4a and 4b and Figures 1a and 1b show the total inventory costs function for those four different values of q_{1k} and $E(L_{11})$. When the decision rule is satisfied, the total inventory costs decrease with a larger volume of transshipment from the other wholesaler. The upper two total inventory costs functions of C_{144} and C_{122} represent the scenarios in which transshipments should not be implemented by the wholesaler because the decision rule is not satisfied. Implementing LT in these two scenarios will only increase the total inventory costs further. Note that similar conclusion can be drawn for both the without spoilage costs and with spoilage costs cases since their shapes and relative distances remain the same. However, the total inventory costs for the cases with spoilage costs with an assumed 30% spoilage ratio are approximately \$500 higher than that for the without spoilage costs due to the higher volume of LT and supplier order required to compensate for the wastage.

Detailed value of the total inventory costs with different size of transshipment for the above four cases under both the without and with spoilage scenarios are presented in Tables 4a and 4b.

Furthermore, the trigger point for implementing LT to minimise the total inventory costs can vary with respect to different spoilage percentages. Assume the expected lead time is $E(L_{ij}) = 3$, the unit purchasing cost from S_j is $p_{ij} = 2.2$, the backordering cost $b = 2$, holding cost $h = 2$. By varying the LT cost q_{ik}, the relationship curves for no spoilage, with spoilage at 1 and 30% are illustrated in Figure 2a. When the decision rule $(1+\theta)(q_{ik} - p_{ij}) - (E(L_{ij})b_i < 0$ is satisfied, LT should be implemented. The corresponding trigger points for the no

Table 3. Decision rule for the regional distribution centre of the case study supermarket chain with one other regional distribution centre and one supplier.

	$q_{11} = 5,\ E(L_{11}) = 3$	$q_{12} = 9,\ E(L_{11}) = 12$	$q_{13} = 6,\ E(L_{11}) = 4$	$q_{14} = 7,\ E(L_{11}) = 1$
Total inventory costs function	C_{111}	C_{122}	C_{133}	C_{144}
Testing the decision rule	$(1+\theta)(q_{12}-p_{11}) < (E(L_{11})b_1$ Decision rule is satisfied when the backordering costs are higher than the spoilage adjusted LT price differential	$(1+\theta)(q_{12}-p_{11}) > (E(L_{11})b_1$ Decision rule is not satisfied when the backordering costs are lower than the spoilage adjusted LT price differential	$(1+\theta)(q_{12}-p_{11}) < (E(L_{11})b_1$ Decision rule is satisfied when the backordering costs are higher than the spoilage adjusted LT price differential	$(1+\theta)(q_{12}-p_{11}) > (E(L_{11})b_1$ Decision rule is not satisfied when the backordering costs are lower than the spoilage adjusted LT price differential
Decision	Quantity transshipped from the other regional distribution centre W_2	No transshipment. All quantity ordered from the supplier S_1	Quantity transshipped from the other regional distribution centre W_2	No transshipment. All quantity ordered from the supplier S_1

Table 4a. Total inventory costs with different size of transshipment under the without spoilage scenario.

Quantity transshipped x	0	10	20	30	40
C_{111}: $q_{11} = 5$, $E(L_{11}) = 3$	3224	3192	3160	3128	3096
C_{122}: $q_{12} = 9$, $E(L_{11}) = 2$	3585	3613	3641	3669	3697
C_{133}: $q_{13} = 6$, $E(L_{11}) = 4$	2950	2908	2866	2824	2782
C_{144}: $q_{14} = 7$, $E(L_{11}) = 1$	4034	4062	4090	4118	4146

Table 4b. Total inventory costs with different size of transshipment under the with spoilage costs = 0.3 scenario.

Quantity transshipped x	0	10	20	30	40
C_{111}: $q_{11} = 5$, $E(L_{11}) = 3$	3774	3750	3726	3703	3679
C_{122}: $q_{12} = 9$, $E(L_{11}) = 2$	4135	4183	4232	4280	4329
C_{133}: $q_{13} = 6$, $E(L_{11}) = 4$	3500	3469	3439	3408	3377
C_{144}: $q_{14} = 7$, $E(L_{11}) = 1$	4584	4626	4669	4711	4753

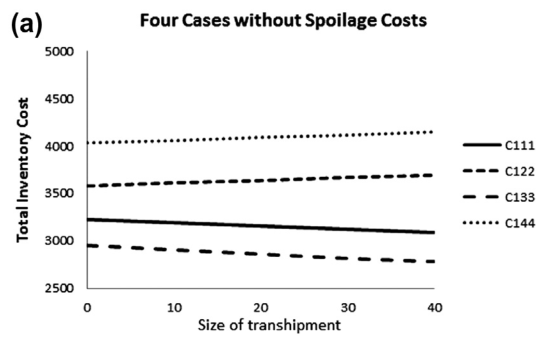

Figure 1a. Total inventory costs with different size of transshipment for four cases having different q_{ik} and $E(L_{ij})$ under the without spoilage scenario.

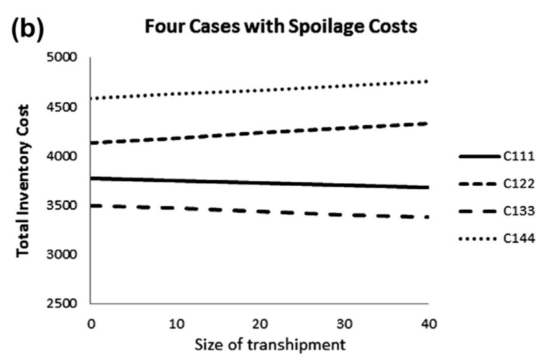

Figure 1b. Total inventory costs with different size of transshipment for four cases having different q_{ik} and $E(L_{ij})$ under the with spoilage scenario.

spoilage case is q_{ik} = \$8.20, higher than that of the 10% spoilage case (\$7.65) and the 30% spoilage case (\$6.82). It reveals that when the spoilage percentage becomes higher, a lower LT cost q_{ik} is required to justify the preventive LT at the start of the scheduling period.

To justify the advantage of the inclusion of spoilage costs, numerical results in Figures 2a and 2b compare the total inventory costs for the inclusion of spoilage costs with that for no inclusion of spoilage costs.

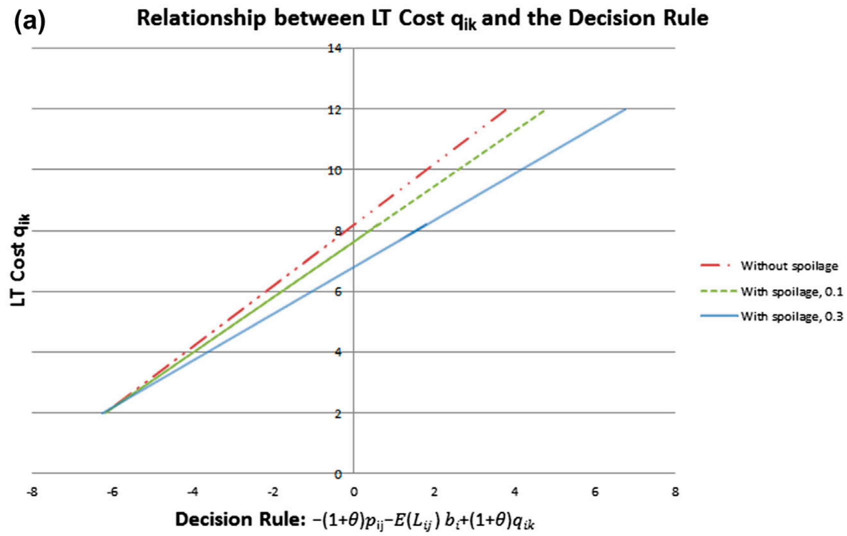

Figure 2a. Comparative relationship between LT cost and the decision rule for given spoilage percentages.

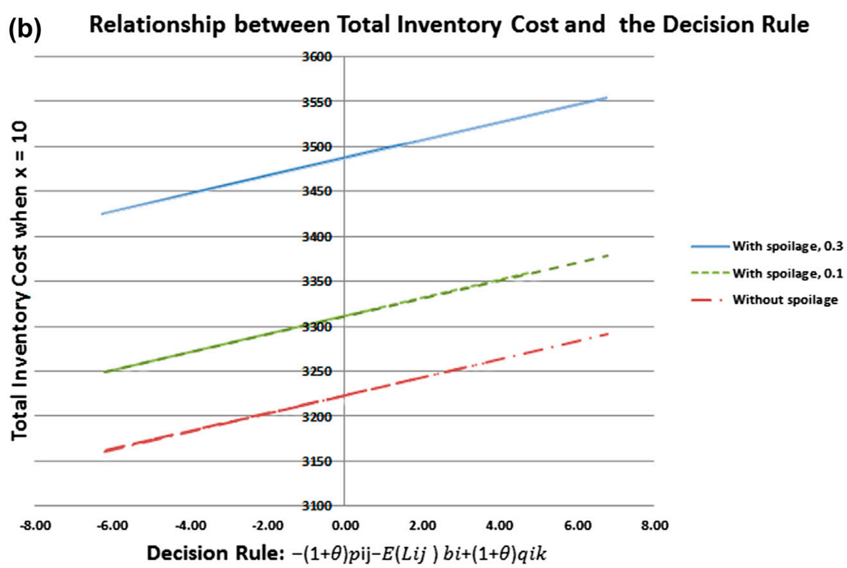

Figure 2b. Comparative relationship between total inventory costs and the decision rule for given spoilage percentages.

When the decision rule $(1+\theta)(q_{ik}-p_{ij})-(E(L_{ij})b_i<0$ is satisfied, LT should be implemented. Assume the size of LT is $x=10$. Under the no spoilage scenario, if q_{ik} falls below \$8.20, it triggers an LT event, with the expected total inventory costs of \$3,224. However, if in fact the spoilage percentage is greater than zero, for example, under the 30% spoilage case, the total inventory costs end up higher at \$3,506, which is 9% higher than the optimal level under the zero spoilage rate scenario.

4.2 *Benefits through application of decision rule*

As indicated in Figures 1a and 1b, the total inventory costs either increase or decrease linearly and monotonically as the size of transshipment varies. It should be noted that the two main parameters determining the direction of the moving trend of the lines are q_{1k} and $E(L_{11})$ as presented in Figures 1a and 1b and Tables 4a and 4b. Once the values of q_{1k} and $E(L_{11})$ are known, logistics managers are able to make the important decision if an LT is justified through the utilisation of the decision rule.

As indicated in Table 3, the total inventory costs functions C_{144} and C_{122} represent the scenarios in which LT should not be recommended if $(1+\theta)(q_{12}-p_{11}) > (E(L_{11})b_1$, i.e. the decision rule is not satisfied when the spoilage adjusted LT price differential is larger than the backordering costs. Under these scenarios, backordering from their usual suppliers to match the stochastic demand is the preferred option among the mix of fast response LT from other warehouses. In this case example, only one supplier and one other regional distribution centre have been considered. However, in real case scenarios, there may be more suppliers and other regional distribution centres. As more choices are available, logistics managers can select among other suppliers and regional distribution centres with different values of q_{1k} and $E(L_{11})$ by assessing their cost differences using the decision rule, aiming at minimising total inventory cost.

The main parameter of the spoilage cost function is the spoilage ratio θ, which is measured from historical retailer orders, sales, procurement and inventory accounting records of this supermarket chain. θ can cover a wide range of values, depending on the category of fresh food. In this case example, three values of θ covering 0, 10 and 30% are considered. Based on the decision rule $-(1+\theta)p_{ij} - E(L_{ij})b_i + (1+\theta)q_{ik} < 0$, as derived in Equation (9), the LT cost q_{ik} needs to be reduced if spoilage ratio θ increases in order to trigger the selection of LT. Figures 2a and 2b present the comparison of these three values of θ, indicating that, for example, when θ is 0.3, the LT cost needs to be reduced from \$8.20 to \$6.82 so as to justify the LT decision. Through utilisation of the decision rule, logistics managers can work out if LT is warranted, given the variation of spoilage ratios in various situations. This is important to achieve the lowest total inventory costs, an ultimate aim of the logistics practitioners. Furthermore, the decision rule provides information regarding the estimated reduction of total inventory costs through LT, as shown in Tables 4a and 4b.

5. Discussion and conclusion

Research studies on LT started in early 1990s (Paterson et al. 2011), and the number of journal articles in this field has grown rapidly over the last five years. However, most of these published LT studies do not have a strong focus on practical application. The International Journal of Production Research has an editorial strategy on knowledge transfer from academics to industry practitioners and the society, and we have designed and structured our study to align with this vision of strong academic/industry link. Our study offers not only a major advance in the existing LT models, but also identifies the spoilage cost as a major factor in the LT model which can undermine the industry practitioners' LT decision to maximise profitability if neglected. Attention to the spoilage cost has been neglected in the existing LT models. Product perishability is one of the primary reasons causing more stock-outs across the supply chain and justifies LT as a managerial response. Previous research investigating the implementation of LT in the fresh food supply chains has been very limited. Our research study aims to develop a more complete LT decision method to assist logistics practitioners in the fresh food supply chains to make cost effective LT decisions in perishable inventory management.

This study is an extension of the LT model (Lau and Nakandala 2012) that has formulated the decision rules for determining whether LT is preferred to effect minimisation of the total inventory costs, and plans to achieve at least two main contributions in the literature and practice. Based on industry feedback from industry practitioners of the case study supermarket chain, in the generalisation and application of the previously developed LT model for non-perishable products (Lau and Nakandala 2012) that minimises the total inventory costs through the trade-off between the four key cost components (purchasing, backordering, holding and LT costs) to a supermarket chain, the intended solution cannot duplicate the expected benefits as postulated. Further investigation of the products and operations of this supermarket chain has identified product perishability as an influencing variable in the total inventory cost function. The LT decision model has thus been enhanced to incorporate the spoilage cost and formulated more accurate LT decision rules to assist logistics practitioners in the fresh food supply chains to support cost effective LT decision that maximises profitability for perishable inventory management. Once the spoilage cost has been integrated into the previously developed LT model, our results demonstrate that perishable products (unlike normal products) with spoilage cost require a lower LT cost to make the LT decision profitable for fulfilling urgent outstanding orders. If this principle is neglected, then the total inventory cost for these perishable products would turn out to be higher than the optimised level as postulated in the previous LT model.

Proper applications of our enhanced LT model can add substantial value with key implications for the global logistics and supply chain management of perishable products in the supermarket chains, as well as in other industries. Furthermore, it also alerts previous LT models documented in the literature, which are predominantly applicable to non-perishable goods, should be amended to represent the spoilage cost as a separate cost component in the total inventory costs function to derive more accurate LT decision rules to assist logistics practitioners with cost-optimised LT decisions. As such, this enhanced model fills an existing research gap by formulating the decision rules on direct transshipment of perishable products between downstream players. These decision rules provide practical decision support for many firms that face inventory management problems with perishable products.

The main advantages of our model include ease of application, simplicity of the rules, less cumbersome data input requirements and therefore making this method easily comprehensible and accessible to purchasing and logistics practitioners. The main decision rule requires only the information of spoilage percentage of the product under investigation, unit purchasing cost from suppliers, unit transshipment cost from another wholesaler, own unit backordering cost and the expected lead time from suppliers. Model calculations can easily be performed using spreadsheet software like MS Excel that is widely used by purchasing and logistics managers.

The known limitations of our study include the approximation of the retailer arrival at the wholesaler with a Compound Poisson probability distribution, of the suggested method of estimating spoilage percentage, calculating backorder and holding costs based on previous cost records and management judgement. We assume that the estimates based on historical records would hold true during the forecasting period. However, the current dynamic business environment may challenge this assumption. Furthermore, we assumed the instant delivery of LT and the total fulfilment of the transshipment order, with no partial orders with multiple wholesalers being considered. However, in practice, LT can take positive lead times and wholesalers can place partial orders of transshipments with multiple wholesalers to mitigate risk. Future research can introduce positive lead times for LT in the model development and partial orders with multiple suppliers and wholesalers to enrich the proposed model of this present study. There are opportunities to integrate the decision rules with business applications like enterprise resource planning systems so that existing transaction records data becomes available for estimating real-time total inventory costs for LT decision.

Disclosure statement

No potential conflict of interest was reported by the authors.

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